



\huge National Autonomous University of Mexico

\huge Faculty of Engineering

Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

Authors:

Dr. Moisés Velasco-Lozano

\huge Representative Elementary Volume

1. Introduction

In petroleum engineering, a porous medium is defined as a system composed of solid rock grains and interconnected pore spaces, with the latter being saturated by one or more fluids, such as oil, water, and/or gas. Within this system, the solid matrix, pore structure, and fluids interact with one another. Consequently, changes in one component may directly influence the behavior and properties of the others.

Understanding the characteristics of porous media is fundamental in reservoir engineering because the storage and transport of reservoir fluids occur within the interconnected pore network of the rock.

Representative Elementary Volume (REV)

Fluid flow through porous media should generally not be analyzed at the microscopic or pore scale when developing conventional reservoir-scale flow models. This is due to the inherent geometrical and structural complexity of the pore system. In particular:

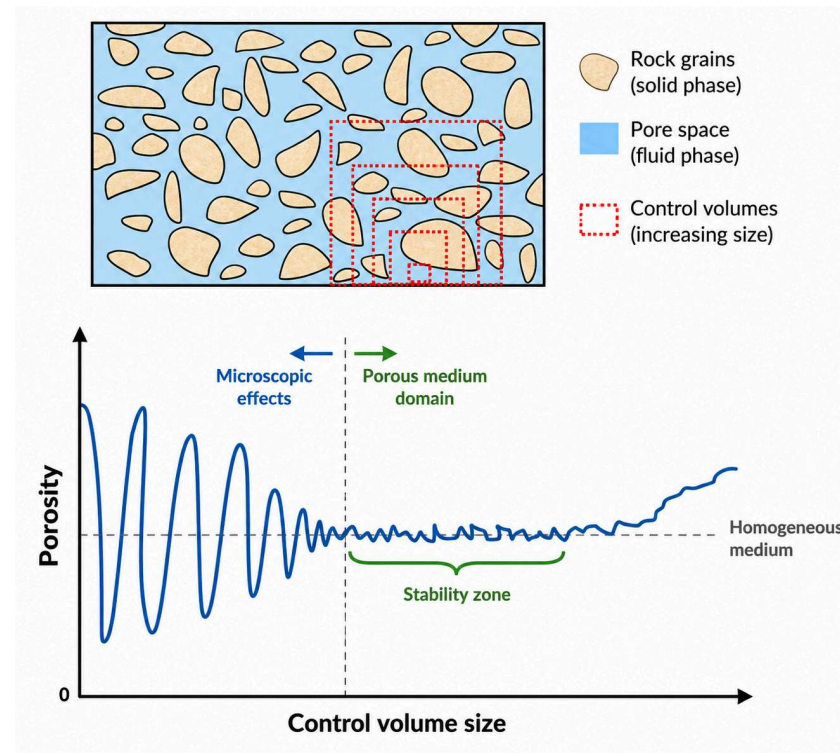
- Pores cannot be represented simply as straight, uniform tubes.
- Pore spaces have irregular geometries, variable sizes, and highly tortuous flow paths.
- Connectivity between pores is complex and forms a disordered three-dimensional network.
- A rigorous mathematical description of every individual pore and flow pathway in a reservoir rock is generally impractical.

For these reasons, conventional fluid-flow problems are commonly formulated at a macroscopic scale, where the detailed pore-scale structure is represented through averaged properties such as porosity, permeability, and fluid saturation. This approach requires defining an appropriate volume of porous material that is sufficiently large to provide representative average properties but sufficiently small to preserve the local characteristics

of the reservoir. This volume is known as the Representative Elementary Volume (REV).

The selection of the REV involves an important balance. If the selected volume is too small, calculated properties such as porosity may fluctuate considerably because individual pores and rock grains dominate the measurement. Conversely, if the volume is too large, the calculated property may average different geological features and therefore fail to represent the local conditions surrounding the point of interest. As the size of the sampling volume increases, fluctuations in the measured property generally decrease until a range is reached in which the average value becomes approximately stable. This range defines the REV, over which continuum-scale properties can be meaningfully assigned to the porous medium.

Thus, the REV provides the conceptual connection between the complex microscopic pore structure and the macroscopic continuum description used in reservoir engineering:



The negative sign indicates that fluid naturally flows in the direction of decreasing hydraulic potential or hydraulic head.

Darcy's experiments demonstrated that the ability of a fluid to move through a porous material depends not only on the applied driving force but also on the properties of the porous medium and the fluid itself. This concept was later generalized for petroleum-reservoir applications by introducing permeability, k , and fluid viscosity, μ , allowing Darcy's Law to be expressed in terms of the pressure gradient.

Today, Darcy's Law provides the foundation for describing single-phase and multiphase flow through reservoir rocks and is extensively used to model the movement of oil, water, and gas in subsurface formations.

2. Numerical solution

2.1 Import libraries

```
In [17]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [18]: # =====
# 1. Generate a synthetic 2D porous medium
# =====

np.random.seed(10)
nx = 200
ny = 200

# Target porosity
phi_target = 0.10

# Binary medium:
# 1 = pore
# 0 = solid
medium = (np.random.rand(nx, ny) < phi_target).astype(int)

# =====
# 2. Define the center of the control volumes
# =====

cx = nx // 2
cy = ny // 2

# =====
# 3. Increase control-volume size and calculate porosity
# =====

sizes = np.arange(5, 151, 2)

porosity = []

for size in sizes:

    half = size // 2

    x1 = max(cx - half, 0)
    x2 = min(cx + half, nx)

    y1 = max(cy - half, 0)
    y2 = min(cy + half, ny)

    control_volume = medium[x1:x2, y1:y2]

    phi = np.mean(control_volume)

    porosity.append(phi)

porosity = np.array(porosity)
```

2.3 Calculation and plotting

```
In [19]: # =====
# 4. Plot porous medium and REV behavior
# =====

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# -----
# Porous medium
# -----

axes[0].imshow(medium, cmap='gray_r', origin='lower')
```

```

axes[0].set_title('Synthetic Porous Medium')
axes[0].set_xlabel('x')
axes[0].set_ylabel('y')

# -----
# Porosity versus control-volume size
# -----

axes[1].plot(
    sizes,
    porosity,
    linewidth=2,
    color='red'
)

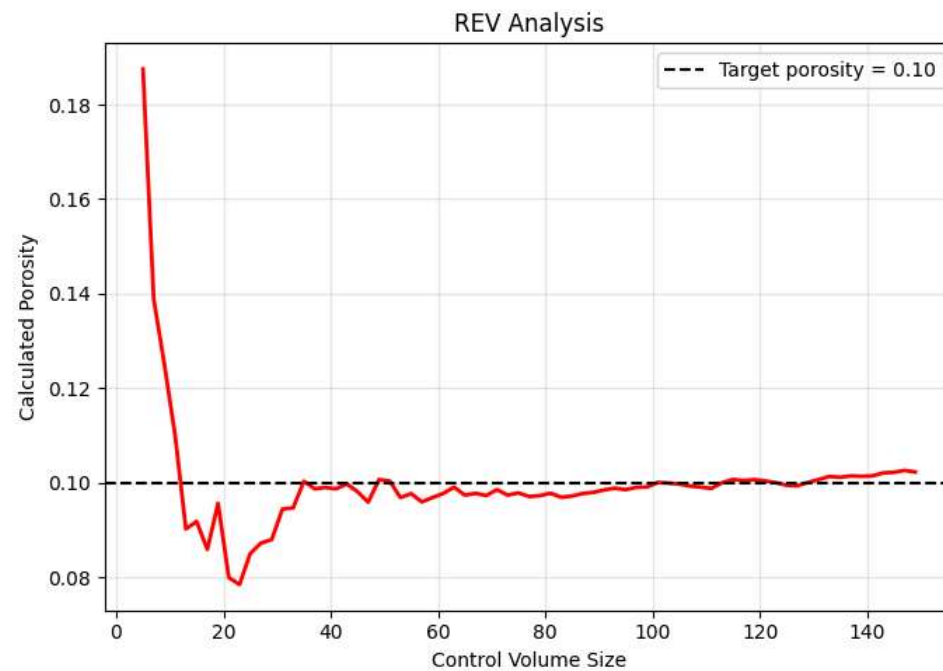
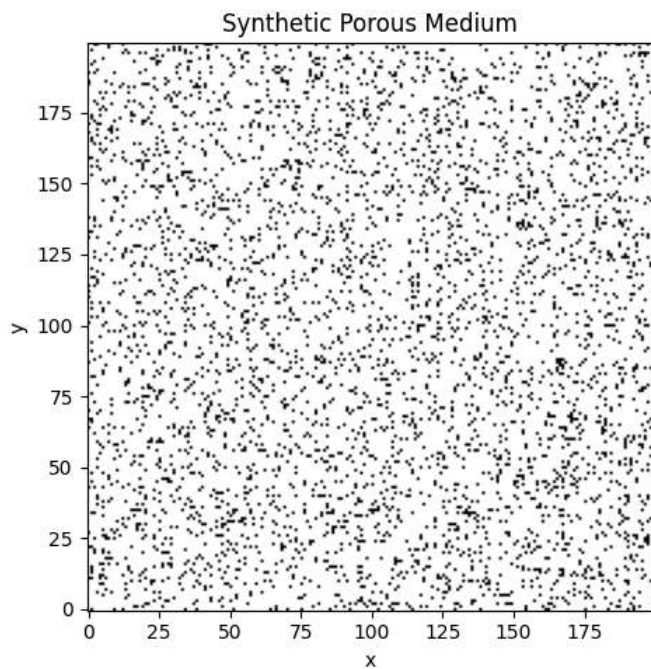
axes[1].axhline(
    phi_target,
    linestyle='--',
    color='black',
    label=f'Target porosity = {phi_target:.2f}'
)

axes[1].set_xlabel('Control Volume Size')
axes[1].set_ylabel('Calculated Porosity')
axes[1].set_title('REV Analysis')

axes[1].grid(alpha=0.3)
axes[1].legend()

plt.tight_layout()
plt.show()

```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

Dr. Moisés Velasco-Lozano

Associate Professor
Petroleum Engineering Department
National Autonomous University of Mexico (UNAM)

mvelasco@ingenieria.unam.edu
mvelasco@unam.mx

In []: